Magnetoelastic Waves in Ferromagnetic Thin Films Mediated by Dipolar Interactions: Supplementary data

Hiroki Yoshida[1], Ryohei Kono[1], Manato Fujimoto[2], Motoki Asano[3], Daiki Hatanaka[3], Kei Yamamoto[4,5], Shuichi Murakami[2,5,6]

[1] *Department of Physics, Institute of Science Tokyo, 2-12-1 Ookayama, Meguro-ku, Tokyo 152-8551, Japan*
[2] *Department of Applied Physics, The University of Tokyo, 7-3-1 Hongo, Bunkyo-ku, Tokyo 113-8656, Japan*
[3] *Basic Research Laboratories, NTT, Inc., 3-1 Morinosato-Wakamiya, Atsugi-shi, Kanagawa 243-0198, Japan*
[4] *Advanced Science Research Center, Japan Atomic Energy Agency, Tokai, Ibaraki 319-1195, Japan*
[5] *Center for Emergent Matter Science, RIKEN, 2-1 Hirosawa, Wako, Saitama 351-0198, Japan*
[6] *International Institute for Sustainability with Knotted Chiral Meta Matter (WPI-SKCM²), Hiroshima University, Higashi-hiroshima, Hiroshima 739-0046, Japan*

1 Jacobian matrix

Here we calculate the Jacobian matrix and its determinant for the transformation $\boldsymbol{r} \mapsto \boldsymbol{R} = \boldsymbol{r} + \boldsymbol{u}(\boldsymbol{r}, t)$ up to first order in $\boldsymbol{u}$.

The definition of the Jacobian matrix J is

$$J_{ij}(\boldsymbol{r}) \coloneqq \delta_{ij} + \frac{\partial u_i}{\partial r_j}. \tag{1}$$

Then, its determinant is given as

$$\begin{aligned}
\det J &= \varepsilon_{i_1 i_2 i_3} J_{1i_1} J_{2i_2} J_{3i_3} \\
&= \varepsilon_{i_1 i_2 i_3} \left(\delta_{1i_1} + \frac{\partial u_1}{\partial r_{i_1}} \right) \left(\delta_{2i_2} + \frac{\partial u_2}{\partial r_{i_2}} \right) \left(\delta_{3i_3} + \frac{\partial u_3}{\partial r_{i_3}} \right) \\
&= 1 + \frac{\partial u_1}{\partial r_1} + \frac{\partial u_2}{\partial r_2} + \frac{\partial u_3}{\partial r_3} + \mathcal{O}\left((\partial u)^2 \right) \\
&\approx 1 + \nabla \cdot \boldsymbol{u}.
\end{aligned} \tag{2}$$

For a nonsingular 3×3 matrix A, its inverse is given by

$$A^{-1} = \frac{1}{\det A} C^T, \tag{3}$$

where C is a cofactor matrix defined by

$$C_{ij} \coloneqq \frac{1}{2} \varepsilon_{i i_1 i_2} \varepsilon_{j j_1 j_2} A_{i_1 j_1} A_{i_2 j_2}. \tag{4}$$

Applying this to the Jacobian matrix, we obtain

$$\begin{aligned}\left[\left(J^{-1}\right)^T\right]_{ij} &= \left(J^{-1}\right)_{ji} \\ &= \frac{1}{\det J} C_{ij} \\ &= \frac{1}{2\det J}\varepsilon_{ii_1i_2}\varepsilon_{jj_1j_2}J_{i_1j_1}J_{i_2j_2} \\ &= \frac{1}{2\det J}\varepsilon_{ii_1i_2}\varepsilon_{jj_1j_2}\left(\delta_{i_1j_1}+\frac{\partial u_{i_1}}{\partial r_{j_1}}\right)\left(\delta_{i_2j_2}+\frac{\partial u_{i_2}}{\partial r_{j_2}}\right) \\ &\approx \frac{1}{2}\left(1-\nabla\cdot\boldsymbol{u}\right)\left(\varepsilon_{ii_1i_2}\varepsilon_{ji_1i_2}+\varepsilon_{ij_1i_2}\varepsilon_{jj_1j_2}\frac{u_{i_2}}{r_{j_2}}+\varepsilon_{ii_1j_2}\varepsilon_{jj_1j_2}\frac{u_{i_1}}{r_{j_1}}\right) \\ &= \left(1-\frac{\partial u_{i_1}}{\partial r_{i_1}}\right)\delta_{ij}+\frac{1}{2}\left(1-\frac{\partial u_{i_1}}{\partial r_{i_1}}\right)\left(2\delta_{ij}\frac{\partial u_{i_1}}{\partial r_{i_1}}-2\frac{\partial u_j}{\partial r_i}\right) \\ &\approx \delta_{ij}-\frac{\partial u_j}{\partial r_i}. \end{aligned} \tag{5}$$

2 Derivation of the dipolar field

Here, we derive the dipolar field that gives rise to magnetoelastic coupling.

We start from Maxwell's equations in the Lagrangian coordinate $\boldsymbol{R}$:

$$\nabla_{\boldsymbol{R}}\cdot\left(\boldsymbol{H}_{\text{in}}(\boldsymbol{R},t)+\boldsymbol{M}(\boldsymbol{R},t)\right)=0, \tag{6}$$

$$\nabla_{\boldsymbol{R}}\times\boldsymbol{H}_{\text{in}}(\boldsymbol{R},t)\approx 0, \tag{7}$$

where $\nabla_{\boldsymbol{R}}$ denotes the derivative with respect to $\boldsymbol{R}$. We then rewrite these equations in terms of the Eulerian coordinate $\boldsymbol{r}$. In doing so, we must take into account the conservation of magnetization [1]:

$$\begin{aligned} \int_{V_R}\mathrm{d}^3R\,\boldsymbol{M}(\boldsymbol{R},t) &= \int_{V_r}\mathrm{d}^3r\,\boldsymbol{M}(\boldsymbol{r},t) \\ \Rightarrow \int_{V_r}\mathrm{d}^3r\,(\det J)\boldsymbol{M}(\boldsymbol{R}(\boldsymbol{r}),t) &= \int_{V_r}\mathrm{d}^3r\,\boldsymbol{M}(\boldsymbol{r},t) \qquad \forall V_r. \\ \therefore \boldsymbol{M}(\boldsymbol{R}) &= \frac{1}{\det J(\boldsymbol{r})}\boldsymbol{M}(\boldsymbol{r}). \end{aligned} \tag{8}$$

Then, using the Jacobian matrix of the coordinate transformation, the Maxwell's equations become

$$\begin{aligned} &\nabla_{\boldsymbol{R}}\cdot\left(\boldsymbol{H}_{\text{in}}(\boldsymbol{R},t)+\boldsymbol{M}(\boldsymbol{R},t)\right)=0 \\ \Rightarrow &\left[\left(J^{-1}\right)^T\right]_{ij}\frac{\partial}{\partial r_j}\left(h_i(\boldsymbol{r},t)+\frac{1}{\det J}M_i(\boldsymbol{r})\right)=0, \end{aligned} \tag{9}$$

and

$$\begin{aligned} &\nabla_{\boldsymbol{R}}\times\boldsymbol{H}_{\text{in}}(\boldsymbol{R},t)\approx 0 \\ \Rightarrow &\,\varepsilon_{ijk}\left[\left(J^{-1}\right)^T\right]_{il}\frac{\partial}{\partial r_l}h_j(\boldsymbol{r})\approx 0 \qquad \forall k=x,y,z. \end{aligned} \tag{10}$$

From the results of Sec. 1, up to first order in $\boldsymbol{m}$ and $\boldsymbol{u}$, we obtain

$$\nabla_r\cdot\left(\boldsymbol{h}^{\text{in}}(\boldsymbol{r})+\boldsymbol{m}(\boldsymbol{r})\right)=\boldsymbol{M}_0\cdot\nabla_r\left(\nabla_r\cdot\boldsymbol{u}(\boldsymbol{r})\right), \tag{11}$$

$$\nabla_r\times\boldsymbol{h}^{\text{in}}(\boldsymbol{r})=0, \tag{12}$$

for points inside the thin film. The equations outside of the film are the same as those in the previous section:

$$\nabla\cdot\boldsymbol{h}^{\text{out}}(\boldsymbol{r},t)=0,$$

$$\nabla\times\boldsymbol{h}^{\text{out}}(\boldsymbol{r},t)\approx 0.$$

Since $\boldsymbol{h}^{\text{in/out}}$ are rotation-free even in the presence of magnetoelastic coupling under the magnetostatic approximation, we introduce scalar potential as

$$\boldsymbol{h}^{\text{in/out}} = -\nabla\phi^{\text{in/out}}.$$

Outside the film, the solution is given by

$$\phi^{\text{out}} = \begin{cases} Ae^{+kz} & (z < -\frac{L}{2}) \\ Be^{-kz} & (z > +\frac{L}{2}), \end{cases} \tag{13}$$

because $\phi^{\text{out}} \to 0$ as $|z| \to \infty$.

The boundary conditions are expressed in terms of the continuity of the tangential component of the magnetic field and the normal component of the magnetic flux density:

$$\boldsymbol{H}^{\text{in}} \cdot \boldsymbol{t} = \boldsymbol{H}^{\text{out}} \cdot \boldsymbol{t}, \tag{14}$$

$$\left(\boldsymbol{H}^{\text{in}} + \boldsymbol{M}\right) \cdot \boldsymbol{n} = \boldsymbol{H}^{\text{out}} \cdot \boldsymbol{n}, \tag{15}$$

where $\boldsymbol{t}$ and $\boldsymbol{n}$ are the tangential and normal vectors to the surface, respectively. Since the system is uniform in the y direction, we restrict our analysis to the xz plane. In this case, these vectors are given by

$$\boldsymbol{t} = \left(1 + \frac{\partial u_x}{\partial x}, 0, \frac{\partial u_z}{\partial x}\right), \tag{16}$$

$$\boldsymbol{n} = \left(-\frac{\partial u_z}{\partial x}, 0, 1 + \frac{\partial u_x}{\partial x}\right). \tag{17}$$

Using the solution (13), we obtain the following boundary condition for ϕ^{in}:

$$\left(\pm k + \partial_z\right)\phi^{\text{in}}|_{z=\pm\frac{L}{2}} = \left(m_z - ikM_{0,x}u_z\right)|_{z=\pm\frac{L}{2}}, \tag{18}$$

where the signs correspond to the boundaries at $z = \pm L/2$. Accordingly, we need to solve the differential equation

$$\nabla^2\phi^{\text{in}} = \nabla \cdot \boldsymbol{m} - \boldsymbol{M}_0 \cdot \nabla\left(\nabla \cdot \boldsymbol{u}(\boldsymbol{r})\right), \tag{19}$$

subject to the boundary condition (18).

To this end, we introduce a Green's function $g(z, z')$ satisfying

$$\nabla^2 g(z, z') = \delta(z - z'), \qquad \left(z, z' \in \left(-\frac{L}{2}, \frac{L}{2}\right)\right), \tag{20}$$

with the boundary condition

$$\left(\pm k + \partial_z\right) g(z, z')|_{z=\pm\frac{L}{2}} = 0. \tag{21}$$

Using this $g(z, z')$, ϕ^{in} can be written as

$$\phi^{\text{in}}(z) = \int \mathrm{d}z' \left[\nabla g(z, z') \cdot \boldsymbol{m}(z') - \nabla\left(\boldsymbol{M}_0 \cdot \nabla g(z, z')\right) \cdot \boldsymbol{u}(z')\right]. \tag{22}$$

The solution of Eqs. (20) and (21) is obtained, for example, by Fourier transformation as

$$g(z, z') = -\frac{1}{2k}e^{-k|z-z'|}. \tag{23}$$

Taking the gradient, we obtain the dipolar field (as given in the main text):

$$\boldsymbol{h}^{\text{in}}(z) = \int_{-\frac{L}{2}}^{\frac{L}{2}} \mathrm{d}z' \left[G^m(z, z')\boldsymbol{m}(z') + G^u(z, z')\boldsymbol{u}(z')\right], \tag{24}$$

where the first term is the same as that in Ref. [2], and

$$G^m(z, z') = \begin{pmatrix} -G_P(z, z') & 0 & -iG_Q(z, z') \\ 0 & 0 & 0 \\ -iG_Q(z, z') & 0 & G_{P'}(z, z') \end{pmatrix}, \tag{25}$$

$$G^u(z, z') = -ikM_{0,x}G^m(z, z'), \tag{26}$$

$$G_P(z, z') = \frac{k}{2}e^{-k|z-z'|}, \tag{27}$$

$$G_Q(z, z') = \operatorname{sgn}(z - z')G_P(z, z'), \tag{28}$$

$$G_{P'}(z, z') = G_P(z, z') - \delta(z - z'). \tag{29}$$

3 Derivation of the force

Here, we derive the force induced by magnetoelastic coupling.

The Euler–Lagrange equation with respect to u_i^* yields

$$\int \mathrm{d}z \Re \left[\rho \ddot{u}_i - \left(-\mu_0 \frac{\delta h_a^*}{\delta u_i^*} h_a + \partial_i \sigma_{ij} \right) \right] = 0, \tag{30}$$

where the second term represents the volume force arising from the dipolar field.

A straightforward calculation shows that

$$\int_{-\infty}^{\infty} \mathrm{d}z'' \left(G^m(z'', z) \right)^* G^m(z'', z') = \int_{-\infty}^{\infty} \mathrm{d}z'' G^m(z, z'') G^m(z'', z') = -G^m(z, z'), \tag{31}$$

$$\int_{-\infty}^{\infty} \mathrm{d}z'' \left(G^m(z'', z) \right)^* G^u(z'', z') = \int_{-\infty}^{\infty} \mathrm{d}z'' G^m(z, z'') G^u(z'', z') = -G^u(z, z'). \tag{32}$$

Then,

$$\begin{aligned} \int \mathrm{d}z \frac{\delta h_a^*(z)}{\delta u_i^*} h_a(z) &= \int \mathrm{d}z \mathrm{d}z' \mathrm{d}z'' (+ikM_{0,x}) \left(G^m(z, z') \right)_{ia}^* \left[G^m(z, z'') \boldsymbol{m}(z'') + G^u(z, z'') \boldsymbol{u}(z'') \right]_a \\ &= -ikM_{0,x} \int \mathrm{d}z' \int \mathrm{d}z'' \left[G^m(z', z'') \boldsymbol{m}(z'') + G^u(z', z'') \boldsymbol{u}(z'') \right]_i \\ &= -ikM_{0,x} \int \mathrm{d}z' h_i^{\mathrm{in}}(z'). \end{aligned} \tag{33}$$

Thus, we obtain

$$f_i(z) :== -ik\mu_0 M_{0,x} h_i(z). \tag{34}$$

References

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